Practice Set A

Class 12 Mathematics, Chapter 5: Continuity and Differentiability

Maximum marks: **51** **20** questions

Name _____ Date _____ Score _____ / 51

1. All questions are compulsory. Marks are shown against each question.
2. Show every step of working: steps carry marks, not just the final answer.
3. Where a question offers OR, attempt only one of the two choices.

Q No.	Question	Marks

Level 1

8 marks

1 MCQ, CBSE 2024 (65/4/1) — 1 mark

For what value of k is the function $f(x) = \begin{cases} \dfrac{\sqrt{4+x}-2}{x}, & x \neq 0 \\ k, & x = 0 \end{cases}$ continuous at $x = 0$?

(A) 0

(B) $\dfrac{1}{4}$

(C) 1

(D) 4

2 MCQ, CBSE 2024 (65/1/1) — 1 mark

Which of the following statements is true for the function $f(x) = \begin{cases} x^2+3, & x \neq 0 \\ 1, & x = 0 \end{cases}$?

(A) $f(x)$ is continuous and differentiable $\forall\, x \in \mathbb{R}$

(B) $f(x)$ is continuous $\forall\, x \in \mathbb{R}$

(C) $f(x)$ is continuous and differentiable $\forall\, x \in \mathbb{R} - \{0\}$

(D) $f(x)$ is discontinuous at infinitely many points

3 MCQ, CBSE 2024 (65/4/1) — 1 mark

Derivative of e^{2x} with respect to e^x is:

(A) e^x **(B)** $2e^x$

(C) $2e^{2x}$ **(D)** $2e^{3x}$

4 MCQ, CBSE 2024 (65/1/1) — 1 mark

The derivative of $\sin(x^2)$ w.r.t. x, at $x = \sqrt{\pi}$ is:

(A) 1 **(B)** -1

(C) $-2\sqrt{\pi}$ **(D)** $2\sqrt{\pi}$

Q No.	Question	Marks
5	MCQ, CBSE 2024 (65/3/1) The derivative of 2^x w.r.t. 3^x is: **(A)** $\left(\frac{3}{2}\right)^x \frac{\log 2}{\log 3}$ **(B)** $\left(\frac{2}{3}\right)^x \frac{\log 3}{\log 2}$ **(C)** $\left(\frac{2}{3}\right)^x \frac{\log 2}{\log 3}$ **(D)** $\left(\frac{3}{2}\right)^x \frac{\log 3}{\log 2}$	1 mark
6	Assertion-Reason, CBSE SQP 2024-25 **Assertion (A):** Consider the function defined as $f(x) = \|x\| + \|x-1\|, x \in \mathbb{R}$. Then $f(x)$ is not differentiable at $x = 0$ and $x = 1$. **Reason (R):** Suppose f be defined and continuous on (a, b) and $c \in (a, b)$; then $f(x)$ is not differentiable at $x = c$ if $\lim_{h\to 0^-} \frac{f(c+h)-f(c)}{h} \neq \lim_{h\to 0^+} \frac{f(c+h)-f(c)}{h}$. **(A)** Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A). **(B)** Both Assertion (A) and Reason (R) are true, but Reason (R) is **not** the correct explanation of the Assertion (A). **(C)** Assertion (A) is true, but Reason (R) is false. **(D)** Assertion (A) is false, but Reason (R) is true.	1 mark
7	Short answer, CBSE 2024 (65/2/1) If $y = \operatorname{cosec}(\cot^{-1} x)$, then prove that $\sqrt{1+x^2}\frac{dy}{dx} - x = 0$.	2 marks

Level 2

18 marks

8	Short answer, PYQ-style Find the value of k for which the function $f(x) = \begin{cases} \frac{k\cos x}{\pi - 2x}, & x \neq \frac{\pi}{2} \\ 5, & x = \frac{\pi}{2} \end{cases}$ is continuous at $x = \frac{\pi}{2}$.	2 marks
9	Short answer, CBSE SQP 2026-27 If $y = x^{\log x} + (\log x)^x$, $x > 1$, then evaluate $\frac{dy}{dx}$.	3 marks
10	Short answer, CBSE 2023 (65/1/3) If $x = \sqrt{a^{\tan^{-1} t}}$, $y = \sqrt{a^{\cot^{-1} t}}$, then show that $x\frac{dy}{dx} + y = 0$.	2 marks
11	Short answer, CBSE 2024 (65(B)) If $x = a\cos\theta$ and $y = b\sin\theta$, then prove that $\frac{d^2y}{dx^2} = -\frac{b^4}{a^2y^3}$.	3 marks

Q No.	Question	Marks
12	Short answer, CBSE 2024 (65/3/1) If $x\cos(p+y)+\cos p\sin(p+y)=0$, prove that $\cos p\,\dfrac{dy}{dx}=-\cos^2(p+y)$, where p is a constant.	3 marks
13	Short answer, CBSE SQP 2026-27 If $y=\sin(x+y)$ and $\dfrac{dy}{dx}=0$, then prove that $y=\pm 1$.	2 marks
14	Short answer, CBSE 2024 (65/3/1) Find the value of a and b so that the function f defined as $f(x)=\begin{cases}\dfrac{x-2}{\lvert x-2\rvert}+a, & \text{if } x<2\\ a+b, & \text{if } x=2\\ \dfrac{x-2}{\lvert x-2\rvert}+b, & \text{if } x>2\end{cases}$ is a continuous function.	3 marks

Level 3 — 25 marks

15	Short answer, PYQ-style If $y=\left(x+\sqrt{1+x^2}\right)^n$, show that $(1+x^2)\dfrac{d^2y}{dx^2}+x\dfrac{dy}{dx}-n^2y=0$.	3 marks
16	Long answer, PYQ-style If $(a+bx)e^{y/x}=x$, prove that $x^3\dfrac{d^2y}{dx^2}=\left(x\dfrac{dy}{dx}-y\right)^2$.	5 marks
17	Short answer, CBSE 2024 (65/2/2) If $x^{30}y^{20}=(x+y)^{50}$, prove that $\dfrac{dy}{dx}=\dfrac{y}{x}$.	3 marks
18	Long answer, CBSE SQP 2024-25 If $f:\mathbb{R}\to\mathbb{R}$ is defined by $f(x)=\lvert x\rvert^3$, show that $f''(x)$ exists for all real x and find it.	5 marks
19	Long answer, CBSE SQP 2026-27 If $x=\sin^3 t$, $y=\cos^3 t$, evaluate $\dfrac{d^2y}{dx^2}$ at $\left(\dfrac{1}{2\sqrt{2}},\dfrac{1}{2\sqrt{2}}\right)$.	5 marks
20	Case study, CBSE 2023 (65/4/1) **Case Study:** Let $f(x)$ be a real valued function. Then its Left Hand Derivative (L.H.D.) is $Lf'(a)=\lim\limits_{h\to 0}\dfrac{f(a-h)-f(a)}{-h}$ and its Right Hand Derivative (R.H.D.) is $Rf'(a)=\lim\limits_{h\to 0}\dfrac{f(a+h)-f(a)}{h}$. Also, a function $f(x)$ is said to be differentiable at $x=a$ if its L.H.D. and R.H.D. at $x=a$ exist and both are equal. For the function $f(x)=\begin{cases}\lvert x-3\rvert, & x\geq 1\\ \dfrac{x^2}{4}-\dfrac{3x}{2}+\dfrac{13}{4}, & x<1\end{cases}$ answer the following questions: **(i)** What is R.H.D. of $f(x)$ at $x=1$? — 1 mark **(ii)** What is L.H.D. of $f(x)$ at $x=1$? — 1 mark **(iii)** Check if the function $f(x)$ is differentiable at $x=1$. — 2 marks	4 marks

Answer key

Chapter 5: Continuity and Differentiability, Practice Set A. Full step-wise solutions are in the solutions sheet.

Level 1

1	(B) $\dfrac{1}{4}$
2	(C) $f(x)$ is continuous and differentiable $\forall\, x \in \mathbb{R} - \{0\}$
3	(B) $2e^x$
4	(C) $-2\sqrt{\pi}$
5	(C) $\left(\dfrac{2}{3}\right)^x \dfrac{\log 2}{\log 3}$
6	(A) Both A and R are true and R is the correct explanation of A
7	$\sqrt{1+x^2}\,\dfrac{dy}{dx} - x = 0$ (proved)

Level 2

8	$k = 10$
9	$\dfrac{dy}{dx} = x^{\log x} \cdot \dfrac{2\log x}{x} + (\log x)^x \left(\log(\log x) + \dfrac{1}{\log x}\right)$
10	$x\dfrac{dy}{dx} + y = 0$ (proved)
11	$\dfrac{d^2y}{dx^2} = -\dfrac{b^4}{a^2y^3}$ (proved)
12	$\cos p\,\dfrac{dy}{dx} = -\cos^2(p+y)$ (proved)
13	$y = \pm 1$ (proved)
14	$a = 1, b = -1$

Level 3

15	$(1+x^2)\dfrac{d^2y}{dx^2} + x\dfrac{dy}{dx} - n^2y = 0$ (proved)
16	$x^3\dfrac{d^2y}{dx^2} = \left(x\dfrac{dy}{dx} - y\right)^2$ (proved)
17	$\dfrac{dy}{dx} = \dfrac{y}{x}$ (proved)
18	$f''(x) = \begin{cases} -6x, & x < 0 \\ 6x, & x \geq 0 \end{cases}$, i.e. $f''(x) = 6\lvert x\rvert$ for all real x.
19	$\dfrac{d^2y}{dx^2} = \dfrac{4\sqrt{2}}{3}$ at $\left(\dfrac{1}{2\sqrt{2}}, \dfrac{1}{2\sqrt{2}}\right)$
20	(i) R.H.D. $= -1$; (ii) L.H.D. $= -1$; (iii) Yes, f is differentiable at $x = 1$ and $f'(1) = -1$.